Exercise 11: Flow of energy and Ten percent law

Aim:

To understand the unidirectional flow of energy in an ecosystem and transfer of energy follows the 10% law.

Principle:

The student studies about flow of energy and that only about 10% of energy is made available to the next trophic level. Large amount of energy about 90% is lost at each trophic level in a food chain.

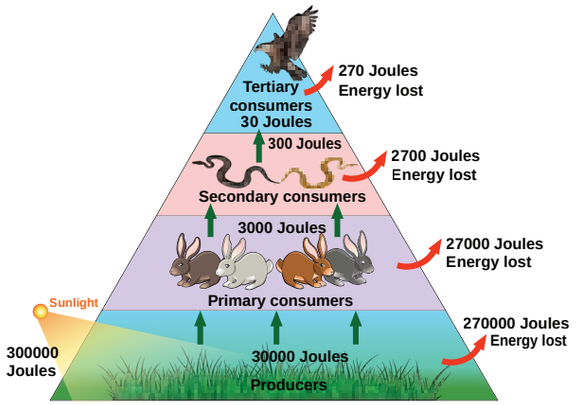


Figure 11: Ten percent law

Requirements:

Problems to be given to students based on different examples with alternating food chain and amount of energy.

The teacher must train the student by giving them various kinds of food chain with different values.

Problem

Analyse the food chain given below and find out the amount of energy received by the organism in third trophic level.

Sun

↓ Grass receives 30,000. J of energy from sun

Grass ⟶ Rabbit ⟶ Snake ⟶ Eagle

Given: The amount of energy in the producers, i.e. grass = 30,000 J.

Solution:

Grass	⟶	Rabbit	⟶	Snake	⟶	Eagle
T_1		T_2		T_3		T_4
Producer		Primary Consumer		Secondary Consumer		Tertiary Consumer

T_1 – Grass (Producer) = 30,000 J of energy

T_2 – Rabbit (Primary Consumer) = ?

T_3 – Snake (Secondary Consumer) = ?

According to the ten percent law, during the transfer of energy, only about 10% of the energy flows from each trophic level to the next lower trophic level. So 10% of energy from T_1 gets transferred to T_2

So T_2 - Rabbit (primary consumer) receives $30000 \times \frac{10}{100} = 3000$ J

Similarly, 10% of energy from T_2 gets transferred to T_3

So T_3 – Snake (Secondary consumer) receives $3000 \times \frac{10}{100} = 300$ J

Answer:

1. The third tropic level T_3 – (Snake) receives 300 J of energy.

TN_GOVT_XII_BIOLOGY_BOTANY_PRACTICAL_MANUAL_PAGES_239-258.indd 234 3/14/2020 12:13:07 PM